

End Behavior and Multiplicity: Sketching Polynomials

Precalculus — Polynomial Function Analysis

End behavior from degree and leading coefficient, zeros and multiplicity, sketching from the features

The two models this sheet uses.

Degree	Leading coefficient	Left end	Right end
even	positive	up	up
even	negative	down	down
odd	positive	down	up
odd	negative	up	down

At a zero, the *multiplicity* of the factor decides the local shape: odd multiplicity means the curve **crosses** the axis there, even multiplicity means it **touches and turns back**. Between consecutive zeros the sign cannot change, so one test value settles a whole interval.

Directions: write end behavior in limit notation, e.g. $f(x) \rightarrow +\infty$ as $x \rightarrow -\infty$. Name the degree and the leading coefficient you used.

1. Describe the end behavior of $f(x) = 2x^3 - 5x + 1$ in both directions. State the degree and the sign of the leading coefficient first, then the two limits.

Answer: _____

2. Describe the end behavior of $g(x) = -x^4 + 3x^2$ in both directions. Does this graph's left end match its right end? Say why in terms of the degree.

Answer: _____

3. The polynomial $f(x) = (x+3)(x-1)^2$ is already factored. List its real zeros, and beside each one state whether the graph crosses the axis or touches and turns there.

Answer: _____

4. Still with $f(x) = (x+3)(x-1)^2$: find $f(0)$, the y -intercept, and find $f(-4)$ to determine the sign of f to the left of its smallest zero. Report both values.

Answer: _____

5. Factor $p(x) = x^3 - 4x^2 + 4x$ completely, then list its zeros. Which zero has multiplicity two, and what does the graph do there?

Answer: _____

6. Describe the end behavior of $h(x) = -3x^5 + 2x^2 - 7$ in both directions. A degree-five polynomial has ends that disagree — the leading coefficient decides which end does what.

Answer: _____

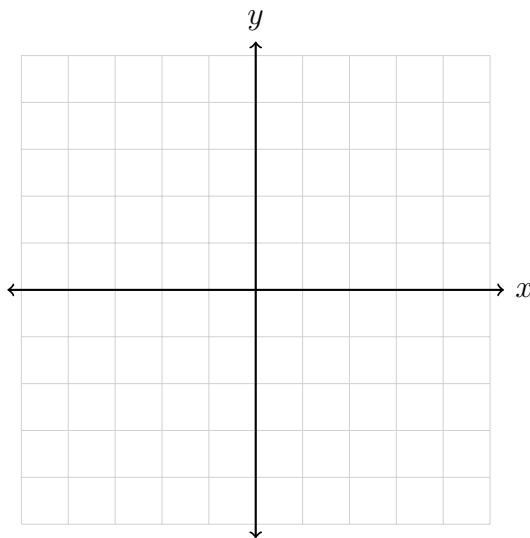
7. For $f(x) = (x + 2)(x - 1)(x - 4)$, evaluate $f(0)$ and $f(2)$. Each value fixes the sign of f on the interval containing that test point; state the two intervals and their signs.

Answer: _____

8. Factor $q(x) = x^4 - 5x^2 + 4$ completely (treat it as a quadratic in x^2 first), then list all four zeros in increasing order. State the multiplicity of each.

Answer: _____

9. Sketch $f(x) = (x + 2)^2(x - 3)$ on the axes below. Before you draw: find $f(0)$, decide the end behavior from the expanded degree and leading coefficient, and mark which zero is a crossing and which is a touch. Label the y -intercept on your sketch.



Answer: _____

10. A student writes: “ $f(x) = -2x^4 + x^3$ has even degree, so both ends rise.” The conclusion is wrong. Explain in one or two sentences which part of the reasoning failed, and give the correct end behavior in both directions.

Answer: _____